Q1. What is the difference between balanced and unbalanced output of a professional audio amplifier and public address system?

Abhijit Dasgupta

A1. Noise-free, clarity and fidelity are some important features of any audio system. Noise can come from a variety of electrical and radio interferences, but most commonly comes from power cables. Humming sound is an example of unwanted noise in an audio system, if there are power cables near the cable carrying unbalanced audio.

Some lighting systems, spotlights or dimmers can also add signal interference in the audio system. There are two types of audio cable connections: unbalanced and balanced.

Unbalanced. An audio cable with two wires—signal and ground—carries unbalanced signal. Here, the ground wire acts as the reference point for the signal. However, it acts like an antenna and picks up unwanted noise along the way. That is, the unbalanced cable carries unwanted noise as well as audio signal along the cable till its end, which is what causes noise in the output (Fig. 1).

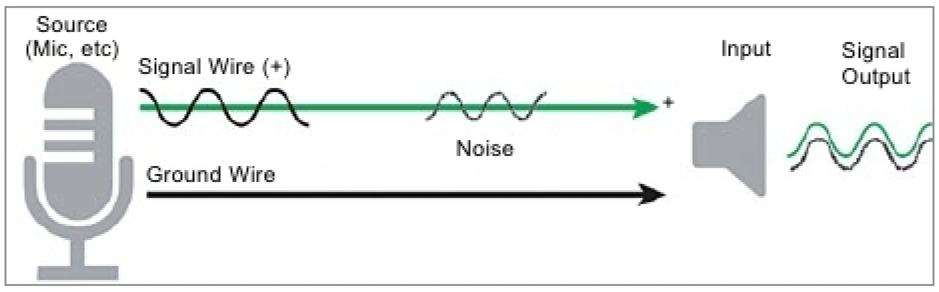

Fig. 1: Unbalanced system (Credit: www.boxcast.com)

Examples of unbalanced cable types are:

TS cable. This is a two-wire cable mostly used at home for short-distance mono audio connections. A 3.5mm TS cable (Fig. 2) is commonly used to connect an audio amplifier.

Fig. 2: Types of audio cables

TS connectors have two contact points separated by an insulator ring—tip and sleeve. The audio signal travels over the tip while the ground uses the sleeve.

RCA cable. This is a type of electrical connector commonly used to carry audio and video signals. The name RCA is derived from Radio Corp. of America. This type of cable has a standard plug on each end, consisting of a central male connector, surrounded by a ring.

RCA audio cables are for unbalanced analogue stereo audio connections over right (red) and left channels (white or black). Because these are unbalanced, maximum cable length is about 7m, or even less.

Unbalanced connections are usually adequate for short-distance home audio systems. But if you need longer distance audio setup, or are working with very-low-level audio recording, balanced connections provide high-gain and noise-free signals.

Balanced. A balanced audio cable has a ground wire along with two copies of the same incoming audio signals—hot (positive) and cold (negative). Hot and cold signals are reversed in polarities (Fig. 3). As these travel down the cable, these cancel each other out. Once these signals reach the other end of the cable, polarity of the cold signal is flipped, so that both signals are in phase and perfectly in sync at final signal output (speaker).

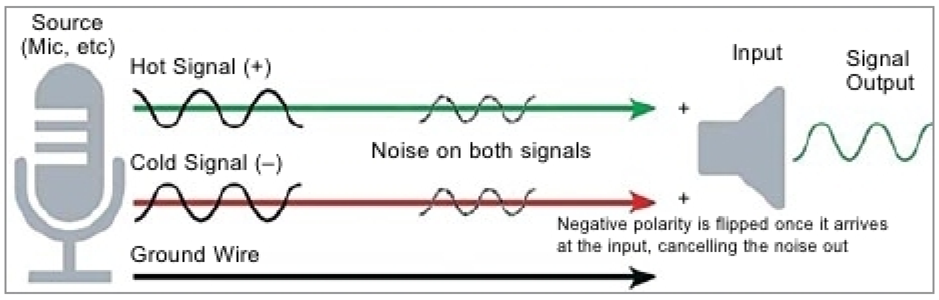

Fig. 3: Balanced system (Credit: www.boxcast.com)

Noise added to both hot and cold signals is not reversed in polarity. So, when the cold signal flips in polarity, noise carried along with the cold signal cancels out with noise in the hot signal. This process is called common-mode rejection. Because balanced signals consist of two in-phase signals, these are also louder (roughly 6dB to 10dB) than unbalanced signals.

Q2. What is the name of the adhesive tape used in SMPS transformer? Suggest some online/offline vendors for the same?

A. Samiuddhin

A2. It is basically a flame-retardant polyester tape normally used in electrical transformers for insulation and other similar purposes. Various types of polyester film acrylic adhesive tapes are available for use in coil wrappings, wire harnesses, motor windings, barriers insulation for keeping distance between coils and bobbins of transformers, and other similar applications. The most common flame-retardant tape available is rated UL 510, UL temperature rated 130°C and UL746A CTI≥600V.

3M is a popular brand available online on Amazon, Mouser and other stores.

Fig. 4: Typical flame-retardant adhesive tapes (Credit: www.nitto.com)

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